



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2002

MATHEMATICS

EXTENSION 2

9:00am – 12:05 pm
Thursday 29th August 2002

General Instructions

- Reading time : 5 minutes
- Working time: 3 hours
- Write using blue or black pen
- *Write your name on each answer booklet*
- Board approved calculators may be used
- A table of standard integrals is provided

- Total Marks (120)
- Attempt Questions 1 – 8
- All questions are of equal value

Students are reminded that this is a trial examination only and cannot in any way guarantee the content or the format of the 2002 Mathematics Extension 2 Higher School Certificate examination

Question 1 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) The complex number u is given by $(-1 + i\sqrt{3})$.
- Show that $u^2 = 2\bar{u}$. 2
 - Evaluate $|u|$ and $\arg u$. 2
 - Show that u is a root of the equation $u^3 - 8 = 0$. 1
- b) If $z = x + iy$ sketch, on separate axes, the locus of z satisfying
- $\operatorname{Re}(z) = |z|$. 2
 - Both $\operatorname{Im}(z) \geq 2$ and $|z - 1| \leq 3$. 3
- c) Given that both c and d are real numbers, find their values such that
- $$\frac{c}{1+i} - \frac{d}{1+2i} = 1.$$
- 2
- d) The points P, Q, R and S on an Argand diagram represent the complex numbers a, b, c and d respectively. 3
- If $a + c = b + d$ and $a - c = i(b - d)$, find what type of quadrilateral $PQRS$ is.

Question 2 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) Sketch the following, showing all essential features.
- $y = \ln x^2$ 2
 - $\sin(x + y) = 1$ 2
 - $y = e^x - e^{-x}$ 2
- b) (i) Draw (without using the Calculus) a neat sketch of the curve
- $$y = x^3 - c^2x; \text{ where } c \text{ is a positive constant.}$$
- Mark clearly any intercepts. 2
- (ii) Use your graph in part (i) to draw neat sketches, on separate number planes, of:
- $y = \frac{1}{x^3 - c^2x}$ 2
 - $y = \left| \frac{1}{x^3 - c^2x} \right|$ 2
 - $y^2 = \frac{1}{x^3 - c^2x}$ 3

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

a) Find the following indefinite integrals.

(i) $\int 2^{2x} dx$ 2

(ii) $\int x e^x dx$ 2

(iii) $\int \frac{2x}{(x+1)(x+3)} dx$ 3

b) By using the substitution $u = t - 4$ evaluate $\int_4^{4.5} \frac{dt}{(t-3)(5-t)}$ 3

c) (i) If $u_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$, $n \geq 2$, 3

prove that $u_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1)u_{n-2}$

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} x^2 \sin x dx$ 2

Question 4 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) For the hyperbola $\frac{x^2}{20} - \frac{y^2}{5} = 1$, find
- (i) the co-ordinates of the two foci, 2
 - (ii) the equations of the asymptotes 2
- b) Explain why $\frac{x^2}{h-19} + \frac{y^2}{3-h} = 1$ cannot represent the equation of an ellipse. 2
- c) Tangents to the ellipse with the equation $x^2 + 4y^2 = 4$ at the points $A(2\cos\theta, \sin\theta)$ and $B(2\cos\alpha, \sin\alpha)$ are at right angles to each other. Show that: $4\tan\theta \cdot \tan\alpha = -1$. 3
- d) A and B are variable points on the rectangular hyperbola $xy = c^2$

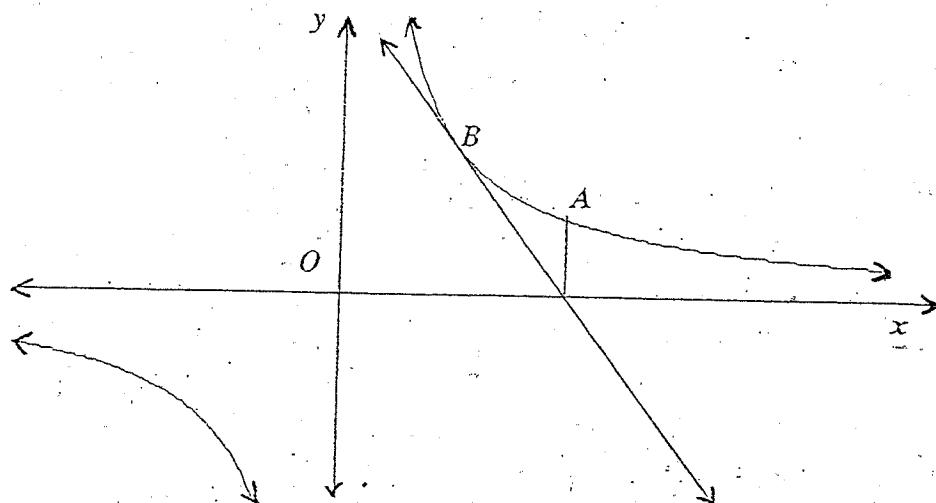


Diagram not to scale.

- (i) The tangent at B passes through the foot of the ordinate of A .
If A and B have parameters t_1 and t_2 , show that $t_1 = 2t_2$. 4
- (ii) Hence prove that the locus of the midpoint of AB is a rectangular hyperbola. 2

Question 5 (15 marks) Use a SEPARATE writing booklet.

- a) Prove that both 1 and -1 are zeroes of multiplicity 2 of the polynomial

$$P(x) = x^6 - 3x^2 + 2.$$

Hence express $P(x)$ as a product of irreducible factors over the field of

- | | | |
|------|---|---|
| (i) | real numbers | 4 |
| (ii) | complex numbers. | 1 |
| b) | (i) Assuming the result $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and using the substitution $x = \cos\theta$ solve the equation $8x^3 - 6x + 1 = 0$. | 3 |
| | (ii) Hence prove that : | |
| | (α) $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0$ | 2 |
| | (β) $\sec \frac{2\pi}{9} + \sec \frac{4\pi}{9} + \sec \frac{8\pi}{9} = 6$ | 2 |
| c) | If α and $-\alpha$ are both roots of $x^3 + mx^2 + nx + h = 0$, show that $mn - h = 0$. | 3 |

Question 6 (15 marks) Use a SEPARATE writing booklet.

Marks

- a) The base of a solid is a right-angled triangle on the horizontal x - y plane; bounded by the lines $y = 0$, $x = 4$ and $y = x$. Vertical cross-sections of the solid, parallel to the y -axis, are semicircles with their diameter on the base of the solid as shown in the diagram below. Find the volume of the solid.

5

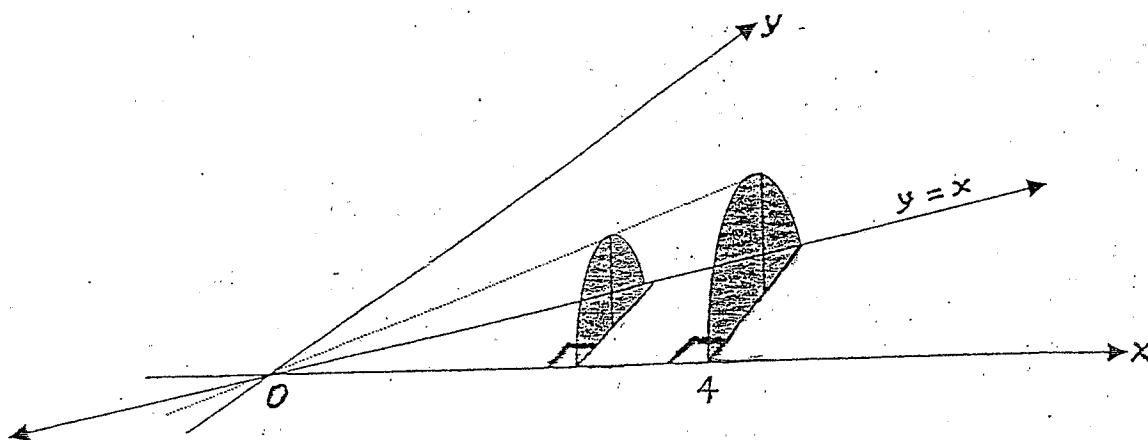


Diagram not to scale

- (b) The area bounded by the line $y = 4 - 2x$, the x -axis and the y -axis, is rotated about the line $x = 4$. By using the method of cylindrical shells find the volume formed.
- (c) Given that for a particular value of x that $\sin^{-1}x$, $\cos^{-1}x$ and $\sin^{-1}(1-x)$ are acute:
- Show that: $\sin(\sin^{-1}x - \cos^{-1}x) = 2x^2 - 1$.
 - Solve the equation: $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(1-x)$.

5

5

Question 7 (15 marks) Use a SEPARATE writing booklet.

- a) A particle is attached to one end of a light string. The other end is fixed. The particle moves in a horizontal circle (below the fixed point) with a speed of 2 m sec^{-1} and the string makes an angle of size $\tan^{-1}(\frac{5}{12})$ with the vertical. Show that the length of the string is approximately 2.5 metres. Take g as 10 m sec^{-2} . 4
- b) A particle of unit mass moves in a straight line with variable acceleration $(\frac{16}{v} - v) \text{ m sec}^{-2}$, where $v \text{ m sec}^{-1}$ is the velocity at time t and $v > 0$, and x is the displacement. If when $t = 0$, $x = 0$ and $v = 2 \text{ m sec}^{-1}$,
- Find an expression for the velocity of the particle at time $t \text{ sec}$. 4
 - Find the limiting velocity of the particle. 2
 - Find the displacement of the particle when $v = 3 \text{ m sec}^{-1}$. 5

Question 8 (15 marks) Use a SEPARATE writing booklet.

Marks

a)

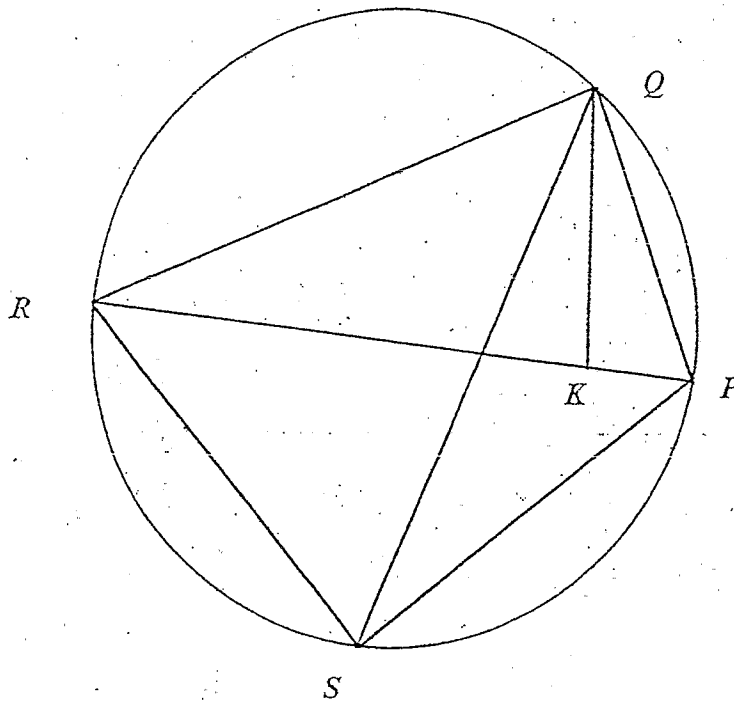


Diagram not to scale.

The above figure is a cyclic quadrilateral. K is the point on RP such that angle PQK is equal to angle SQR .

Let angle $SQR = x^\circ$ and

(i) Show that triangle PQS is similar to triangle KQR and that the triangle PQK is similar to triangle SQR . 6

(ii) Hence show that $PR \cdot SQ = PQ \cdot SR + PS \cdot QR$. 2

b) Prove by Mathematical Induction that: 5

$$\sum_{r=1}^n \sin((2r-1)\theta) = \frac{\sin 2n\theta}{\sin \theta}, \text{ where } n \text{ is a positive integer.}$$

c) For the following statement answer true or false giving a reason for your answer. 2

For $n = 1, 2, 3, \dots$
$$\int_0^1 \frac{dx}{1+x^n} \leq \int_0^1 \frac{dx}{1+x^{n+1}}$$



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Aids To Solutions

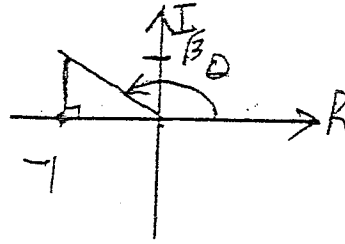
Question 1

(a) $w = -1 + i\sqrt{3}$

(ii) $\arg w = \tan^{-1} \frac{\sqrt{3}}{-1}$

$$= \frac{2\pi}{3}$$

$$|w| = \sqrt{3+1} = 2.$$



(i) $w^2 = \left[2 \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) \right]^2$

$$= 4 \left[\cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right) \right]$$

$$= 4 \left(\cos\left(\frac{\pi}{3}\right) - i \sin\left(\frac{\pi}{3}\right) \right)$$

$$= 4 \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right)$$

$$= 2(-1 - i\sqrt{3})$$

$$= 2\bar{w}$$

Note
This could have been done without mod-arg form.

(iii) $w^3 - 8 = 0$

$$\text{LHS} = \left[2 \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) \right]^3 - 8$$

$$= 8 \left[\cos 2\pi - i \sin 2\pi \right]^3 - 8$$

$$= 8 (1 - i(0)) - 8$$

$$= 8 - 8 = 0$$

= RHS.

(b) $R(z) = |z|$

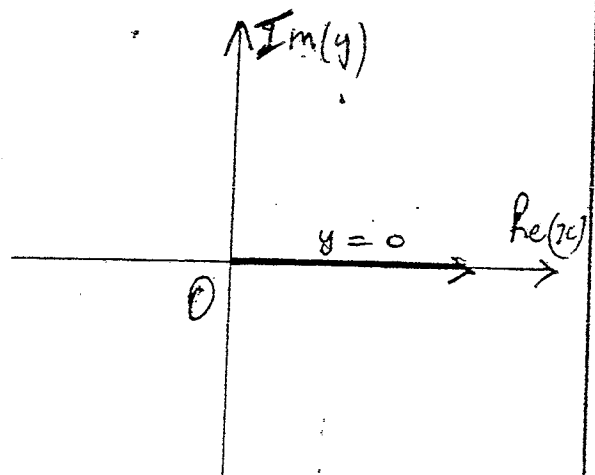
$$x = \sqrt{x^2 + y^2}; \text{ Note } x \geq 0.$$

$$x^2 = x^2 + y^2$$

$$y^2 = 0$$

$$y = 0.$$

$\therefore$ locus is the +ve x-axis and zero



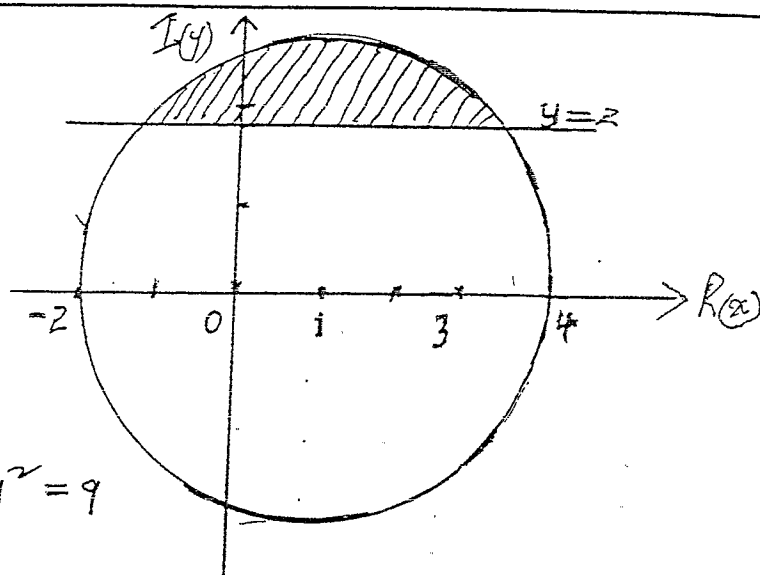
Question 1 (Continued)

b (ii) $\text{Im}(z) \geq 2$

i.e. $y \geq 2$

$|z-1| \leq 3$

circle centre (1,0)
radius = 3



$$(x-1)^2 + y^2 = 9$$

(c) $\frac{c}{1+i} - \frac{d}{1+2i} = 1$

$$\frac{c(1-i)}{1+1} - \frac{d(1-2i)}{1+4} = 1$$

$$\frac{c}{2} - \frac{ci}{2} - \frac{d}{5} + \frac{2di}{5} = 1$$

$$\left(\frac{c}{2} - \frac{d}{5}\right) + i\left(\frac{2d}{5} - \frac{c}{2}\right) = 1 + 0i$$

Equate Real & Im parts.

$$\frac{c}{2} - \frac{d}{5} = 1 \Rightarrow 5c - 2d = 10 \quad \text{--- (1)}$$

$$\frac{2d}{5} - \frac{c}{2} = 0 \Rightarrow 5c - 4d = 0 \quad \text{--- (2)}$$

Solving: $4d - 2d = 10$

$$\left. \begin{array}{l} d = 5 \\ \therefore c = 4 \end{array} \right\}$$

(d) IF $a+c = b+d$

Then $\frac{a+c}{2} = \frac{b+d}{2}$

$\therefore$ midpoint PR = midpoint QS

$\therefore$ diagonals PR and QS bisect each other

IF $a-c = b-d$

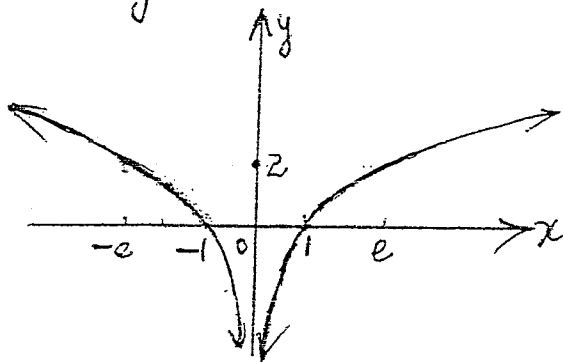
$\therefore$ diagonals PR and QS are equal and perpendicular

$\therefore$ PQRS as a square.

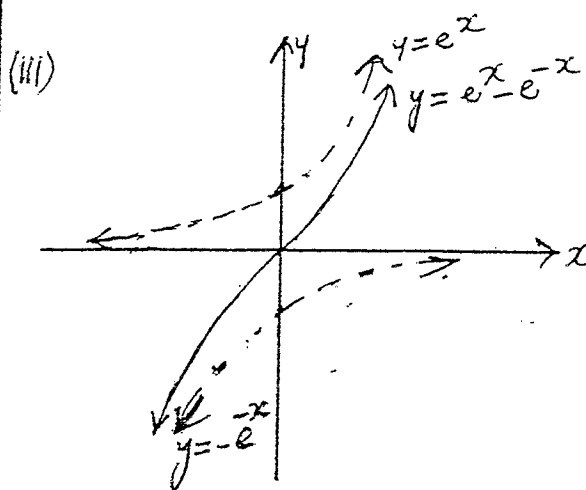
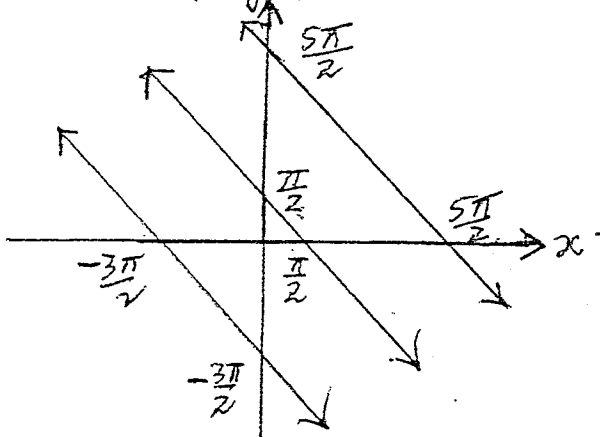
Note A diagram would be useful.

Question 2

(a) (i) $y = \ln x^2$

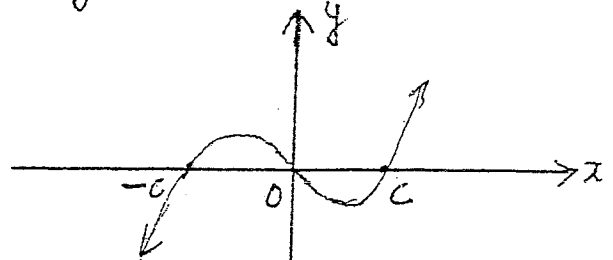


(ii) $\sin(x+y) = 1$



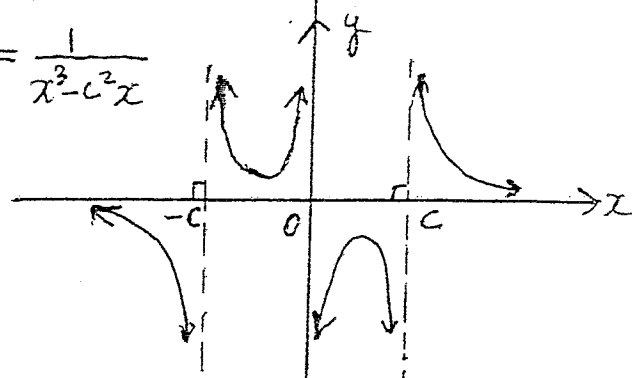
Note In HSC each of these should be about $\frac{1}{2}$ page.

(b) (i) $y = x^3 - c^2x$

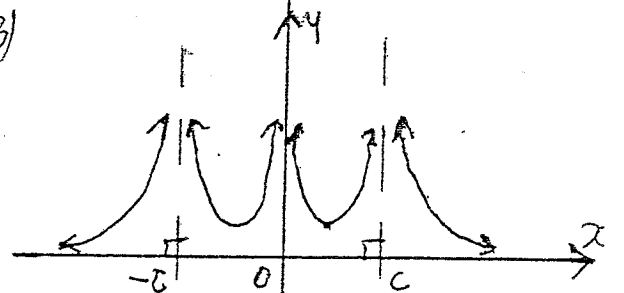


(ii) (a)

$y = \frac{1}{x^3 - c^2x}$

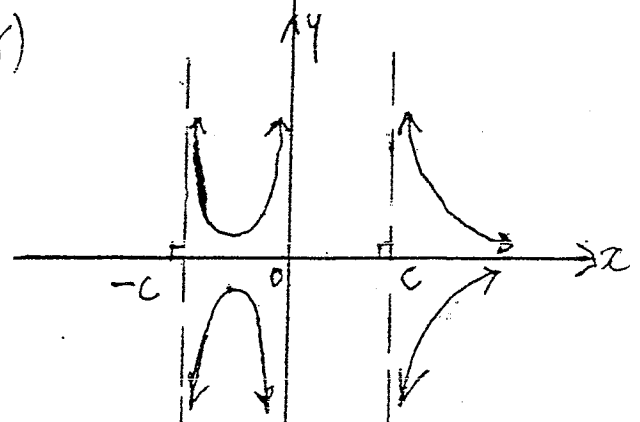


(b)



$y = \frac{1}{|x^3 - c^2x|}$

(c)



$y^2 = \frac{1}{x^3 - c^2x}$

Question 3

$$(a)(i) \int 2^{2x} dx$$

$$= \frac{1}{\ln 4} 2^{2x} + C$$

Note If $y = 2^{2x}$

$$\ln y = 2x \ln 2$$

$$\frac{dy}{dx} = 2 \ln 2 \cdot y$$

$$\frac{dy}{y} = 2 \ln 2 dx$$

$$= 2 \cdot 2^{2x} \ln 2$$

$$= 2^{2x} \ln 4$$

$$(ii) I = \int x e^x dx$$

$$\text{Let } v' = e^x, u = x$$

$$I = x e^x - \int e^x x dx$$

$$= x e^x - e^x + C$$

$$(iii) \frac{2x}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$$

$$\therefore A = -1, B = 3$$

$$I = \int \frac{2x}{(x+1)(x+3)} dx$$

$$= \int \left(-\frac{1}{x+1} + \frac{3}{x+3} \right) dx$$

$$= 3 \ln|x+3| - \ln|x+1| + C$$

$$= \ln \left| \frac{(x+3)^3}{x+1} \right| + C$$

Question 3 (continued)

$$(b) I = \int_4^{4.5} \frac{dt}{(t-3)(5-t)}$$

$$u = t - 4$$

$$du = dt$$

$$t - 3 = u + 1$$

$$5 - t = 1 - u$$

$$\text{when } t = 4, u = 0$$

$$\text{when } t = 4.5, u = \frac{1}{2}$$

$$I = \int_0^{1/2} \frac{du}{1-u^2}$$

$$= \frac{1}{2} \int_0^{1/2} \left(\frac{1}{1-u} + \frac{1}{1+u} \right) du$$

(Note use of partial fractions)

$$= \frac{1}{2} \left[\ln(1+u) - \ln(1-u) \right]_0^{1/2}$$

$$= \frac{1}{2} \left[-\ln\left(\frac{1+u}{1-u}\right) \right]_0^{1/2}$$

$$= \frac{1}{2} \left[\ln\left(\frac{3}{2} \div \frac{1}{2}\right) - \ln 1 \right]$$

$$= \frac{1}{2} \ln 3$$

$$= \ln \sqrt{3}$$

$$(c) (i) U_n = \int_0^{\pi/2} x^n \sin x \, dx$$

$$\text{let } u = x^n, v' = \sin x$$

$$\therefore U_n = \left[-\cos x \cdot x^n \right]_0^{\pi/2} + \int_0^{\pi/2} n x^{n-1} \cos x \, dx$$

$$U_n = n \int_0^{\pi/2} x^{n-1} \cos x \, dx$$

$$\text{Now if } u = x^{n-1}, v' = \cos x$$

$$U_n = n \left\{ \left[x^{n-1} \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} (n-1) x^{n-2} \sin x \, dx \right\}$$

$$U_n = n \left[\left(\frac{\pi}{2} \right)^{n-1} \right] - n(n-1) \int_0^{\pi/2} x^{n-2} \sin x \, dx$$

$$U_n = n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) U_{n-2}$$

$$(ii) \int_0^{\pi/2} x^2 \sin x \, dx$$

$$U_2 = 2 \left(\frac{\pi}{2} \right)^1 - 2(1) U_0$$

$$= \pi - 2 \int_0^{\pi/2} \sin x \, dx$$

$$= \pi - 2 \left[-\cos x \right]_0^{\pi/2}$$

$$= \pi + 2 \left(\cos \frac{\pi}{2} - \cos 0 \right)$$

$$= \pi + 2(0 - 1)$$

$$= \pi - 2$$

Question 14

$$(a) \quad \frac{x^2}{20} - \frac{y^2}{5} =$$

$$a = \sqrt{20} = 2\sqrt{5}$$

$$b = \sqrt{5}$$

$$b^2 = a^2(e^2 - 1)$$

$$\therefore e^2 = 1 + \frac{5}{20}$$

$$e^2 = \frac{5}{4}$$

$$e = \frac{\sqrt{5}}{2}$$

$$(i) \text{ foci : } S = (ae, 0) = (5, 0)$$

$$\text{hence } S' = (-5, 0)$$

$$(ii) \text{ asymptotes : let } \frac{x^2}{20} - \frac{y^2}{5} = 0$$

$$\left(\frac{x}{2\sqrt{5}} + \frac{y}{\sqrt{5}}\right)\left(\frac{x}{2\sqrt{5}} - \frac{y}{\sqrt{5}}\right) = 0$$

$$\text{hence } y = \frac{x}{2} \text{ or } y = -\frac{x}{2}$$

$$(b) \quad \frac{x^2}{h-19} + \frac{y^2}{3-h} = 1$$

$$h-19 > 0 \text{ and } 3-h > 0$$

$$\text{i.e. } h > 19 \text{ and } h < 3$$

sets have no intersection hence not possible.

$$(c) \quad \frac{x^2}{4} + y^2 = 1; \quad a=2, b=1$$

$$\text{Equation of tangent as } \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

$$\text{At A : } \frac{x \cos \theta}{2} + y \sin \theta = 1; \therefore m_1 = -\frac{\cos \theta}{2 \sin \theta}$$

$$\text{At B : } \frac{x \cos \alpha}{2} + y \sin \alpha = 1; \therefore m_2 = -\frac{\cos \alpha}{2 \sin \alpha}$$

$$\left(\frac{-\cos \theta}{2 \sin \theta}\right) \left(\frac{-\cos \alpha}{2 \sin \alpha}\right) = -1$$

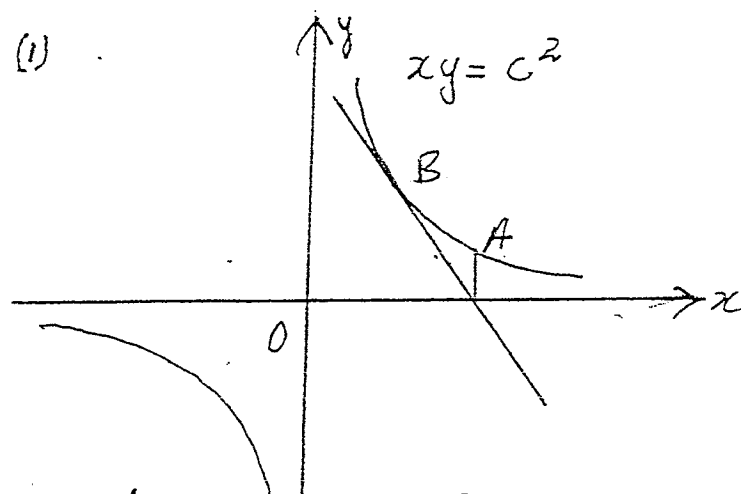
$$\frac{1}{2 \tan \theta} \times \frac{1}{2 \tan \alpha} = -1$$

$$\text{i.e. } 4 \tan \theta \tan \alpha = -1$$

If tangents are at right angles then $m_1 m_2 = -1$

Question 4 (continued)

(d) (i)



Let N = Foot of the ordinate at A.

$$B(c t_2, \frac{c}{t_2}); A(c t_1, \frac{c}{t_1})$$

$$\text{tangent at B: } y' = -\frac{c^2}{x^2}$$

$$\text{gradient m at B is } m = -\frac{c^2}{c^2 t_2^2} = -\frac{1}{t_2^2}$$

$$\text{Eqn: } y - \frac{c}{t_2} = -\frac{1}{t_2^2}(x - c t_2)$$

$$t_2^2 y - c t_2 = -x + c t_2$$

For Co-ordinates of N Let $y = 0$

$$\therefore x = 2c t_2$$

But This is equal to the x co-ordinate of A.

$$c t_1 = 2c t_2$$

$$\therefore t_1 = 2 t_2 \quad \text{--- (1)}$$

(ii) Co-ordinates of the mid point of AB

$$x = \frac{c}{2}(t_1 + t_2) \quad \text{--- (2)}$$

$$y = \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2}\right) \quad \text{--- (3)}$$

Eliminate parameters.

sub (1) in both (2) and (3)

$$x = \frac{c}{2}(3 t_2) \quad \text{--- (2a)}$$

$$y = \left(\frac{1}{2 t_2} + \frac{1}{t_2}\right) \frac{c}{2} \quad \text{---}$$

$$y = \left(\frac{1+2}{2 t_2}\right) \frac{c}{2}$$

$$y = \frac{c}{2}\left(\frac{3}{2 t_2}\right) \quad \text{--- (3a)}$$

(2a) $\times$ (3a)

$$x y = \frac{c}{2}(3 t_2) \times \frac{c}{2}\left(\frac{3}{2 t_2}\right)$$

$$x y = \frac{c^2}{8} \times 9$$

$$8 x y = 9 c^2$$

which is a rectangular hyperbola,

Question 5.

(a) $P(x) = x^6 - 3x^2 + 2$

$$P'(x) = 6x^5 - 6x$$

$$P(1) = 1 - 3 + 2 = 0$$

$$P'(1) = 6 - 6 = 0$$

$\therefore 1$ is a zero of multiplicity 2

$$P(-1) = 1 - 3 + 2 = 0$$

$$P'(-1) = -6 + 6 = 0$$

$\therefore (-1)$ is a zero of multiplicity 2.

Now I.P. $x^2 = t$ we have $t^3 - 3t + 2 = Q(t)$

$$\text{and } Q(-2) = -8 + 6 + 2 = 0$$

$\therefore (t+2)$ is a factor of $Q(t)$

Hence (x^2+2) is a factor of $P(x)$

(i) $P(x) = (x+1)^2(x-1)^2(x^2+2)$

(ii) $P(x) = (x+1)^2(x-1)^2(x+i\sqrt{2})(x-i\sqrt{2})$

(b)(i) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

Now $8x^3 - 6x + 1 = 0$ and $x = \cos\theta$.

$$8\cos^3\theta - 6\cos\theta + 1 = 0$$

$$2(4\cos^3\theta - 3\cos\theta) + 1 = 0$$

$$2\cos 3\theta = -1$$

$$\cos 3\theta = -\frac{1}{2}$$

General soln $3\theta = 2n\pi \pm \cos^{-1}(-\frac{1}{2})$

$$\therefore 3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \dots$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}, \dots$$

$\therefore x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$ 3 unique solutions only.

Question 5 (continued)

(2) $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9}$ represents the sum of the roots

$$\text{sum} = -\frac{b}{a} = \frac{0}{8} = 0$$

$$\therefore \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0.$$

(3) $\sec \frac{2\pi}{9} + \sec \frac{4\pi}{9} + \sec \frac{8\pi}{9}$

$$= \frac{1}{\cos \frac{2\pi}{9}} + \frac{1}{\cos \frac{4\pi}{9}} + \frac{1}{\cos \frac{8\pi}{9}}$$

$$= \frac{\cos \frac{4\pi}{9} \cos \frac{8\pi}{9} + \cos \frac{2\pi}{9} \cos \frac{8\pi}{9} + \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}}{\cos \frac{2\pi}{9} \cos \frac{4\pi}{9} \cos \frac{8\pi}{9}}$$

$$= \frac{\text{sum of roots 2 at a time}}{\text{product of roots}}$$

$$= \frac{-\frac{6}{8}}{-\frac{1}{8}} = 6.$$

(c) $x^3 + mx^2 + nx + h = 0$
let roots be $\alpha, -\alpha, \beta$.

$$\text{sum } \alpha - \alpha + \beta = -m \quad \text{--- (1)}$$

$$\text{sum 2 at a time } \alpha x - \alpha + \alpha x \beta + -\alpha x \beta = n \quad \text{--- (2)}$$

$$\text{product } -\alpha^2 \beta = -h \quad \text{--- (3)}$$

$$\beta = -m \quad \text{--- (1a)}$$

$$-\alpha^2 = n \quad \text{--- (2a)}$$

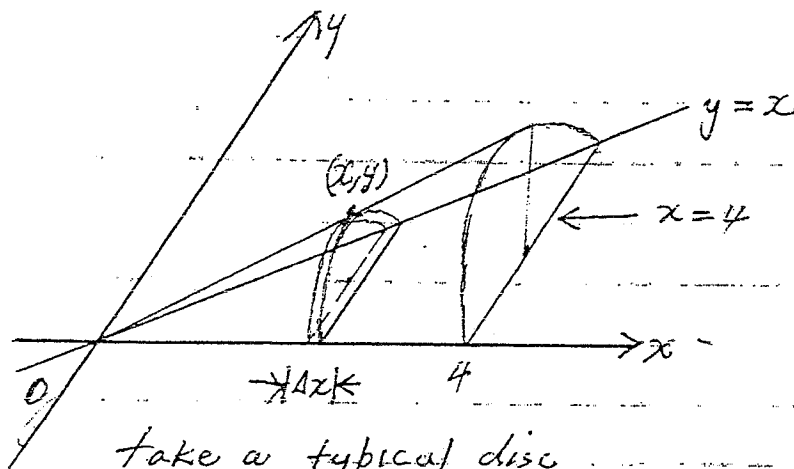
$$\text{Sub in (3)}$$

$$n x - m = -h$$

$$\therefore h - mn = 0$$

Question 6

(a)



take a typical disc

Area of cross-section of semi $\odot$

$$\Delta A = \frac{1}{2} \pi \left(\frac{y}{2} \right)^2$$

$$= \frac{\pi y^2}{8}$$

Volume of slice :

$$\Delta V \doteq \frac{\pi y^2}{8} \cdot \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^4 \frac{\pi y^2}{8} \Delta x$$

$$V = \int_0^4 \frac{\pi y^2}{8} dx$$

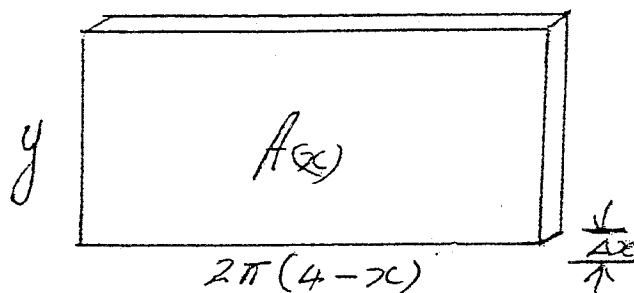
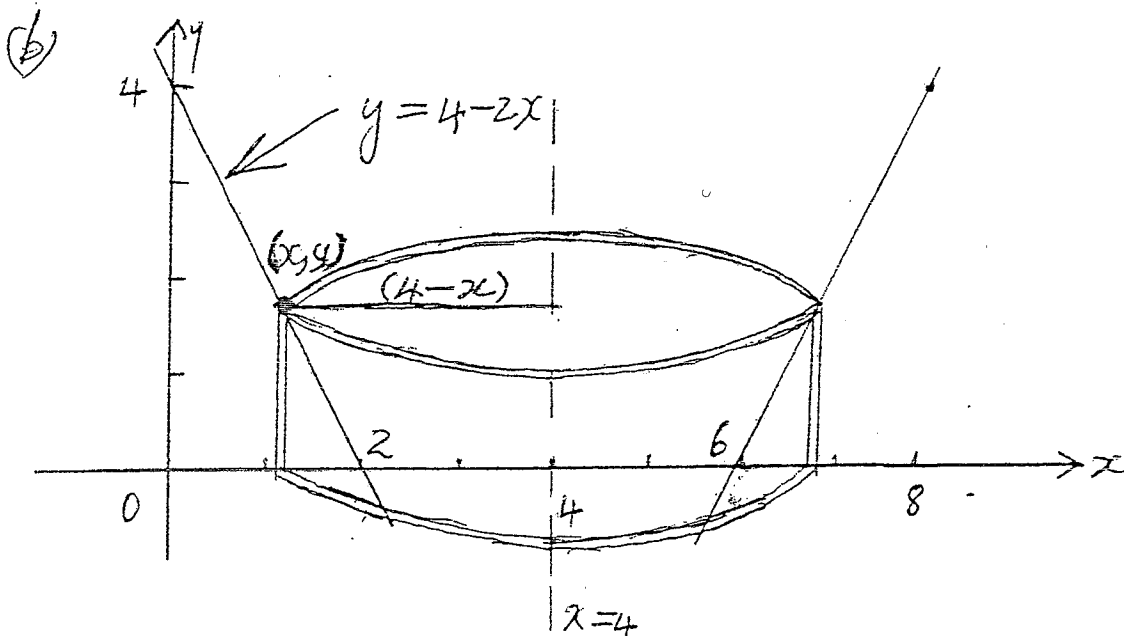
$$V = \frac{\pi}{8} \int_0^4 x^2 dx \quad \text{Note } y=x$$

$$V = \frac{\pi}{8} \left[\frac{x^3}{3} \right]_0^4$$

$$V = \frac{\pi}{8} \times \frac{64}{3}$$

$$\text{Volume} = \frac{8\pi}{3} \text{ units}^3$$

Question 6 (Continued)



$$A(x) = 2\pi(4-x)y$$

$$\Delta V \approx 2\pi(4-x)(4-2x)\Delta x$$

$$V = \lim_{x \rightarrow 0} \sum_{x=0}^2 2\pi(4-x)(4-2x)\Delta x$$

$$V = 2\pi \int_0^2 (16 - 12x + 2x^2) dx$$

$$V = 2\pi \left[16x - 6x^2 + \frac{2x^3}{3} \right]_0^2$$

$$V = 2\pi \left[32 - 24 + \frac{16}{3} \right]$$

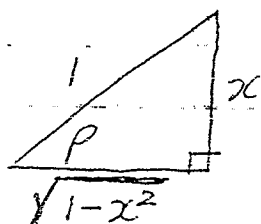
$$V = 2\pi \left[\frac{24}{3} + \frac{16}{3} \right]$$

$$V = \frac{8\pi}{3} \text{ units}^3$$

Question 6 (Continued)

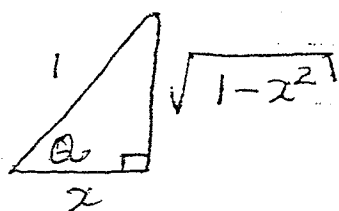
(c)(i) Let $P = \sin^{-1}x$

$\therefore x = \sin P$



Let $Q = \cos^{-1}x$

$\therefore x = \cos Q$



Hence $\cos P = \sqrt{1-x^2}$

and $\sin Q = \sqrt{1-x^2}$

$$\text{LHS} = \sin(\sin^{-1}x - \cos^{-1}x)$$

$$= \sin(P - Q)$$

$$= \sin P \cos Q + \cos P \sin Q$$

$$= x \times x - \sqrt{1-x^2} \times \sqrt{1-x^2}$$

$$= x^2 - (1-x^2)$$

$$= 2x^2 - 1$$

$$= \text{RHS}$$

(ii) $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(1-x)$

$$\sin(\sin^{-1}x - \cos^{-1}x) = 1-x$$

$$2x^2 - 1 = 1-x$$

using result of (i)

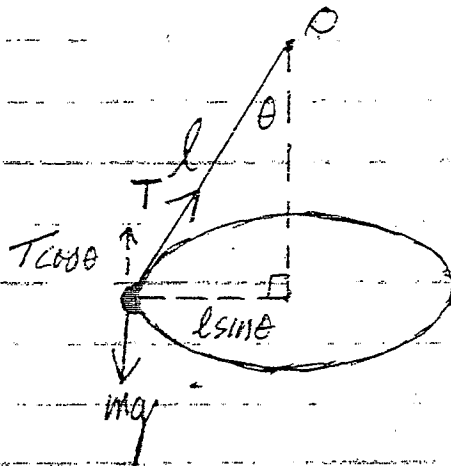
$$2x^2 + x - 2 = 0$$

$$x = \frac{-1 \pm \sqrt{1-4(2x-2)}}{2(2)}$$

$$x = \frac{-1 \pm \sqrt{17}}{4}$$

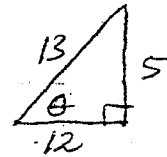
Question 6

(a)



$$\tan \theta = \frac{5}{12}$$

$$\therefore \sin \theta = \frac{5}{13} \text{ using}$$



Resolving Vertically: $T \cos \theta = mg$ — — — (1)

Resolving Horizontally: $T \sin \theta = \frac{mv^2}{r}$ — — — (2)

$$T \sin \theta = \frac{m \times 2^2}{l \sin \theta}$$

$$T \sin \theta = \frac{4m}{l \sin \theta} \text{ — — — (2a)}$$

Now (2a) $\div$ (1)

$$\frac{T \sin \theta}{T \cos \theta} = \frac{4m}{l \sin \theta} \times \frac{1}{mg}$$

$$\tan \theta = \frac{4}{lg \sin \theta}$$

$$l = \frac{4}{(\tan \theta) g \sin \theta}$$

$$l = \frac{4}{\frac{5}{12} \times 10 \times \frac{5}{13}}$$

$$l = \frac{4 \times 12 \times 13}{5 \times 10 \times 5}$$

$$l \approx 2.49 \text{ ie approx } 2.5 \text{ m}$$

Question 7 (continued)

(b) $\ddot{x} = \frac{16}{v} - v$

$$\frac{dv}{dt} = \frac{16}{v} - v$$

$$= \frac{16 - v^2}{v}$$

$$\frac{dt}{dv} = \frac{v}{16 - v^2}$$

$$t = -\frac{1}{2} \ln(16 - v^2) + c$$

when $t=0$, $v=2$.

$$0 = -\frac{1}{2} \ln(16 - 4) + c$$

$$0 = -\frac{1}{2} \ln 12 + c$$

$$c = \frac{1}{2} \ln 12$$

$$t = \frac{1}{2} \ln 12 - \frac{1}{2} \ln(16 - v^2)$$

$$t = \frac{1}{2} \ln \frac{12}{16 - v^2}$$

$$2t = \ln \frac{12}{16 - v^2}$$

$$\frac{12}{16 - v^2} = e^{2t}$$

$$\frac{16 - v^2}{12} = e^{-2t}$$

$$16 - v^2 = 12e^{-2t}$$

$$v^2 = 16 - 12e^{-2t}$$

$$v^2 = 16 \left(1 - \frac{3}{4} e^{-2t}\right)$$

$$v^2 = 16 \left(1 - \frac{3}{4} e^{-2t}\right)$$

$$v = 4 \left(1 - \frac{3}{4} e^{-2t}\right)^{1/2}; v > 0. \quad x = \left[2 \ln\left(\frac{7}{3}\right) - 1\right] \text{ metres.}$$

$$\lim_{t \rightarrow \infty} \left(1 - \frac{3}{4} \cdot \frac{1}{e^{2t}}\right) = 1$$

$$\therefore \lim_{t \rightarrow \infty} \left(1 - \frac{3}{4} \cdot \frac{1}{e^{2t}}\right)^{1/2} = 1$$

$$\therefore \lim_{t \rightarrow \infty} (v) = 4$$

$\therefore$ Limiting velocity = 4 m sec⁻¹

considering $v \frac{dv}{dx} = \frac{16}{v} - v$

$$\frac{dv}{dx} = \frac{16}{v^2} - 1$$

$$\frac{dx}{dv} = \frac{v^2}{16 - v^2}$$

$$\frac{-v^2 + 16}{16} \sqrt{\frac{v^2 + 0}{-v^2 - 16}}$$

Note

$$\frac{dx}{dv} = \frac{16}{16 - v^2} - 1$$

Transform to partial fractions

$$\frac{dx}{dv} = 16 \left[\frac{1}{8(4+v)} + \frac{1}{8(4-v)} \right] - 1$$

$$x = 2 \ln \left| \frac{4+v}{4-v} \right| - v + c$$

when $x=0$, $v=2$.

$$0 = 2 \ln\left(\frac{6}{2}\right) - 2 + c$$

$$c = 2 - 2 \ln 3$$

$$x = 2 \ln \left| \frac{4+v}{4-v} \right| - v + 2 - 2 \ln 3$$

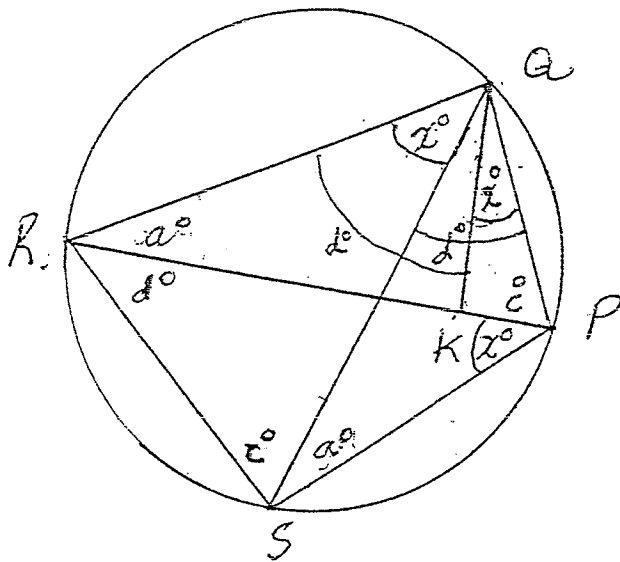
$$x = 2 \ln \left| \frac{4+v}{3(4-v)} \right| + 2 - v$$

when $v=3$

$$x = \left[2 \ln\left(\frac{7}{3}\right) - 1\right] \text{ metres.}$$

Question 8

(a) (i)



Now $\widehat{PAK} = \widehat{SQR} = x^\circ$ (given)
and $\widehat{ASP} = \widehat{QRP} = a^\circ$ (say) $\left(\begin{array}{l} \text{Angles subtended at the circumference} \\ \text{of the circle by a common arc (AP)} \\ \text{are equal} \end{array} \right)$

Similarly $\hat{RPS} = \hat{ROS} = \chi^0$

similarly $\widehat{RSQ} = \widehat{QPR} = 6^\circ$ (say)

similarly $\widehat{SQP} = \widehat{SRP} = d^0(\text{diag})$

$$\therefore \widehat{SAP} = d^s - x^o$$

$$\therefore \hat{R} \hat{a}_k = d^0$$

Now in Δ 's PQS and KQR

$$\hat{P} \hat{Q} S = K \hat{G} R = d^0$$

$$\hat{QSP} = \hat{KRQ} = a^0$$

$$\hat{Q}_{PS} = \hat{Q}_{KR} \text{ (remaining } \Lambda\text{'s of } \Delta \text{ are } =, \text{ because } \Lambda \text{ sum of } \Delta \neq 180^\circ \text{)}^*$$

So Δ s are equiangular

$$\therefore \Delta PQS \parallel \Delta' KAR$$

Also in Δ 's PQR and SQR.

$$\hat{P} \hat{Q} K = S \hat{Q} K = x^0$$

$$\hat{Q}_K = \hat{Q}_R = C^0$$

$$\therefore \widehat{PKQ} = \widehat{SRQ} \text{ (same reason as *)}$$

Δ^1, Δ^2 are equiangular

$\therefore \Delta PQR \sim \Delta SAR$

Question 8 (continued)

(a) (ii) $\Delta PQS \parallel \Delta KQR$

$$\frac{PQ}{KQ} = \frac{PS}{KR} = \frac{QS}{QR} \quad \left(\begin{array}{l} \text{In similar } \Delta\text{'s ratios of corresponding} \\ \text{sides are equal.} \end{array} \right)$$

$$\therefore PS \cdot QR = QS \cdot KR \quad \text{--- --- --- (1)}$$

Also as $\Delta PQR \parallel \Delta SRQ$

$$\frac{PQ}{SQ} = \frac{PR}{SR} = \frac{QR}{QR} \quad \left(\begin{array}{l} \text{In similar } \Delta\text{'s ratios of corresponding} \\ \text{sides are equal.} \end{array} \right)$$

$$PQ \cdot SR = SQ \cdot PR \quad \text{--- --- --- (2)}$$

$$\text{Now } PR \cdot SQ = (PK + KR) \cdot SQ$$

$$= PK \cdot SQ + KR \cdot SQ$$

$$PR \cdot SQ = PQ \cdot SR + PS \cdot QR \quad (\text{using (1) and (2)})$$

Question 8 (continued)

$$b) \sum_{r=1}^n \sin(2r-1)\theta = \frac{\sin^2 n\theta}{\sin \theta}$$

$$\text{Let } T_n = \sin(2n-1)\theta$$

$$\text{Let } S_n = \frac{\sin^2 n\theta}{\sin \theta}$$

$$\sum_{r=1}^n \sin(2r-1)\theta = \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta$$

when $n=1$

$$\frac{\sin^2 n\theta}{\sin \theta} = \frac{\sin^2 \theta}{\sin \theta} = \sin \theta \text{ which is true.}$$

Let $n=k$ be true.

$$\therefore S_k = \frac{\sin^2 k\theta}{\sin \theta}$$

$$\text{Now } S_k + T_{k+1} = \frac{\sin^2 k\theta}{\sin \theta} + \sin[2(k+1)-1]\theta$$

$$= \frac{\sin^2 k\theta}{\sin \theta} + \sin(2k+1)\theta$$

$$= \frac{2\sin^2 k\theta + 2\sin(2k+1)\theta \sin \theta}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + 2\sin \theta [\sin 2k\theta \cos \theta + \cos 2k\theta \sin \theta]}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + \sin 2k\theta \sin 2\theta + 2\sin^2 \theta \cos 2k\theta}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + \sin 2k\theta \sin 2\theta + 2 \times \frac{1}{2} (1 - \cos 2\theta) \cos 2k\theta}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + \sin 2k\theta \sin 2\theta + \cos 2k\theta - \cos 2k\theta \cos 2\theta}{2\sin \theta}$$

$$= \frac{1 - \cos(2k\theta + 2\theta)}{2\sin \theta}$$

$$= \frac{1 - \cos 2(k\theta + \theta)}{2\sin \theta}$$

$$S_k + T_{k+1} = \frac{2\sin^2(k\theta + \theta)}{2\sin \theta} = \frac{\sin^2(k+1)\theta}{\sin \theta}$$

Question 8 (continued)

(b) This result is the required form i.e. S_k with $(k+1)$ in place of k .

The result is true for $n = k+1$ if it is true for $n = k$ i.e. if it is true for one integer then it is true for the next consecutive integer.

∴ true when $n=1$, also true when $n=2$

" ✓ ✓ $n=2$, ✓ ✓ ✓ ✓ $n=3$

" ✓ ✓ $n=3$, ✓ / / / $n=4$ etc

Hence
$$\sum_{r=1}^n \sin (r-1)\theta = \frac{\sin^2 n\theta}{\sin \theta} \text{ true for all}$$

positive integers n .

(c) for $0 \leq x \leq 1$

and $n = 1, 2, 3, \dots$ $x^n \geq x^{n+1}$

$$\therefore 1 + x^n \geq 1 + x^{n+1}$$

$$\therefore \frac{1}{1+x^n} \leq \frac{1}{1+x^{n+1}}$$

$$\therefore \int_0^1 \frac{dx}{1+x^n} \leq \int_0^1 \frac{dx}{1+x^{n+1}}$$

∴ The statement is true.

Question 8 (continued)

(a) (i) $\Delta PQS \parallel \Delta KQR$

$$\frac{PQ}{KQ} = \frac{PS}{KR} = \frac{QS}{QR} \quad \left(\begin{array}{l} \text{In similar } \Delta\text{'s ratio of corresponding} \\ \text{sides are equal.} \end{array} \right)$$

$$\therefore PS \cdot QR = QS \cdot KR \quad \text{--- --- --- (1)}$$

Also as $\Delta PQK \parallel \Delta SQR$

$$\frac{PQ}{SQ} = \frac{PK}{SR} = \frac{QK}{QR} \quad \left(\begin{array}{l} \text{In similar } \Delta\text{'s ratio of corresponding} \\ \text{sides are equal.} \end{array} \right)$$

$$PQ \cdot SR = SQ \cdot PK \quad \text{--- --- --- (2)}$$

$$\text{Now } PR \cdot SQ = (PK + KR) \cdot SQ$$

$$= PK \cdot SQ + KR \cdot SQ.$$

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Question 8 (continued)

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$$\text{Let } T_n = \sin(2n-1)\theta$$

$$\text{Let } S_n = \frac{\sin^2 n\theta}{\sin \theta}$$

$$\sum_{r=1}^n \sin(2r-1)\theta = \sin \theta + \sin 3\theta + \sin 5\theta + \dots + \sin(2n-1)\theta$$

When $n=1$

$$\frac{\sin^2 n\theta}{\sin \theta} = \frac{\sin^2 \theta}{\sin \theta} = \sin \theta \text{ which is true.}$$

Let $n=k$ be true.

$$\therefore S_k = \frac{\sin^2 k\theta}{\sin \theta}$$

$$\text{Now } S_k + T_{k+1} = \frac{\sin^2 k\theta}{\sin \theta} + \sin[2(k+1)-1]\theta$$

$$= \frac{\sin^2 k\theta}{\sin \theta} + \sin(2k+1)\theta$$

$$= \frac{2\sin^2 k\theta + 2\sin(2k+1)\theta \sin \theta}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + 2\sin \theta [\sin 2k\theta \cos \theta + \cos 2k\theta \sin \theta]}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + \sin 2k\theta \sin 2\theta + 2\sin^2 \theta \cos 2k\theta}{2\sin \theta}$$

$$= \frac{1 - \cos 2k\theta + \sin 2k\theta \sin 2\theta + 2 \times \frac{1}{2} (1 - \cos 2\theta) \cos 2k\theta}{2\sin \theta}$$

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$$= \frac{1 - \cos 2(k\theta + \theta)}{2\sin \theta}$$

$$S_k + T_{k+1} = \frac{2\sin^2(k\theta + \theta)}{2\sin \theta} = \frac{\sin^2(k+1)\theta}{\sin \theta}$$

Question 8 (continued)

(b) This result is the required form i.e. S_k with $(k+1)$ in place of k .

The result is true for $n = k+1$ if it is true for $n = k$ i.e. if it is true for one integer then it is true for the next consecutive integer.

• true when $n=1$, also true when $n=2$

" ✓ ✓ $n=2$, ✓ ✓ ✓ ✓ $n=3$

" ✓ ✓ $n=3$, ✓ / / / $n=4$ etc

Hence $\sum_{r=1}^n \sin(r-1)\theta = \frac{\sin^2 n\theta}{\sin \theta}$ true for all positive integers n .

(c) for $0 \leq x \leq 1$

and $n=1, 2, 3, \dots$ $x^n \geq x^{n+1}$

$$\therefore 1+x^n \geq 1+x^{n+1}$$

$$\therefore \frac{1}{1+x^n} \leq \frac{1}{1+x^{n+1}}$$

$$\therefore \int_0^1 \frac{dx}{1+x^n} \leq \int_0^1 \frac{dx}{1+x^{n+1}}$$

$\therefore$ The statement is true.